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Electrospun SiZrOC nanofibers with tunable amorphous-crystalline structures for high-performance electromagnetic absorption

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ABSTRACT

In polymer-derived ceramics, the electronegativity difference between Si and C dictates the covalent bonding topology and electron density distribution, thereby affecting polarization, electron transport, and electromagnetic (EM) wave attenuation. However, the role of amorphous-crystal heterostructures formed by in situ nanocrystal growth on polarization and EM absorption remains poorly understood. Here, SiZrOC nanofibers were fabricated via electrospinning followed by controlled pyrolysis, resulting in ZrO_2 nanocrystal embedded in an amorphous matrix. By modulating the pyrolysis temperature, the crystallization behavior of ZrO_2 nanocrystal and the corresponding dielectric properties were optimized, enabling balanced conductive loss, polarization loss, and multiple scattering. The nanofibers pyrolyzed at 1000 °C delivered exceptionally EM absorption properties, achieving a minimum reflection loss of −78.77 dB at 12.56 GHz with a thickness of 3.16 mm and a maximum effective absorption bandwidth of 6.8 GHz at a thickness of 3.88 mm. These results highlight the potential of SiZrOC nanofibers as lightweight, high-performance absorbers for next-generation wireless communication and high-power electronic devices.

1. Introduction

With the rapid advancement of microwave communication technologies, electromagnetic pollution has become increasingly severe, causing interference to electronic devices and posing risks to human health [1–4]. Microwave absorbing materials (MAMs) are considered an effective solution, as they convert microwave energy into other forms such as thermal energy [5–8]. Various MAMs have been explored, including one-dimensional materials (e.g., SiC nanowires, carbon nanotubes et al.) [9–12], two-dimensional materials (e.g., graphene, MXene et al.) [13–15], and bulk materials (e.g., carbon-based composites, ceramics et al.) [16–18]. However, many MAMs still suffer from narrow effective absorption bandwidths and excessive density, limiting their ability to meet the practical requirements of "thin thickness, low density, strong absorption, wide frequency band" absorption of microwave demands [19,20].

Among these candidates, polymer-derived ceramics (PDCs) have recently attracted considerable attention due to intrinsic advantages, including precursor homogeneity at the molecular level, relative lower processing temperatures compared to conventional ceramic routes, and the feasibility of designing novel compositions with tailored functionalities [21,22]. Nevertheless, conventional PDCs generally exhibit limited absorption bandwidths and poor impedance matching, which results in significant reflection of incident microwaves and ultimately restrict their practical application [23,24]. Recent studies have demonstrated that compositional regulation of PDCs can effectively tune EM parameters and enhanced microwave absorption [25–27]. For example, Li et al. [28] fabricate SiFeBOC ceramics containing Fe_3Si spheres and SiO_2 nanowires via the PDC method, achieving a minimum reflection loss (RL_{min}) of −47.68 dB and an effective absorption bandwidth of 4.39 GHz at only 1.6 mm thickness, owing to the synergistic effects of magnetic-dielectric coupling, interfacial polarization, and improved impedance matching. Chen et al. [29] proposed a controllable phase-transformation strategy in Sm-doped SiBCN ceramics, achieving an RL_{min} of −54.24 dB and a broad bandwidth of 7.48 GHz through the combined effects of wave-transparent Si_3N_4 lossy SiC, and graphite

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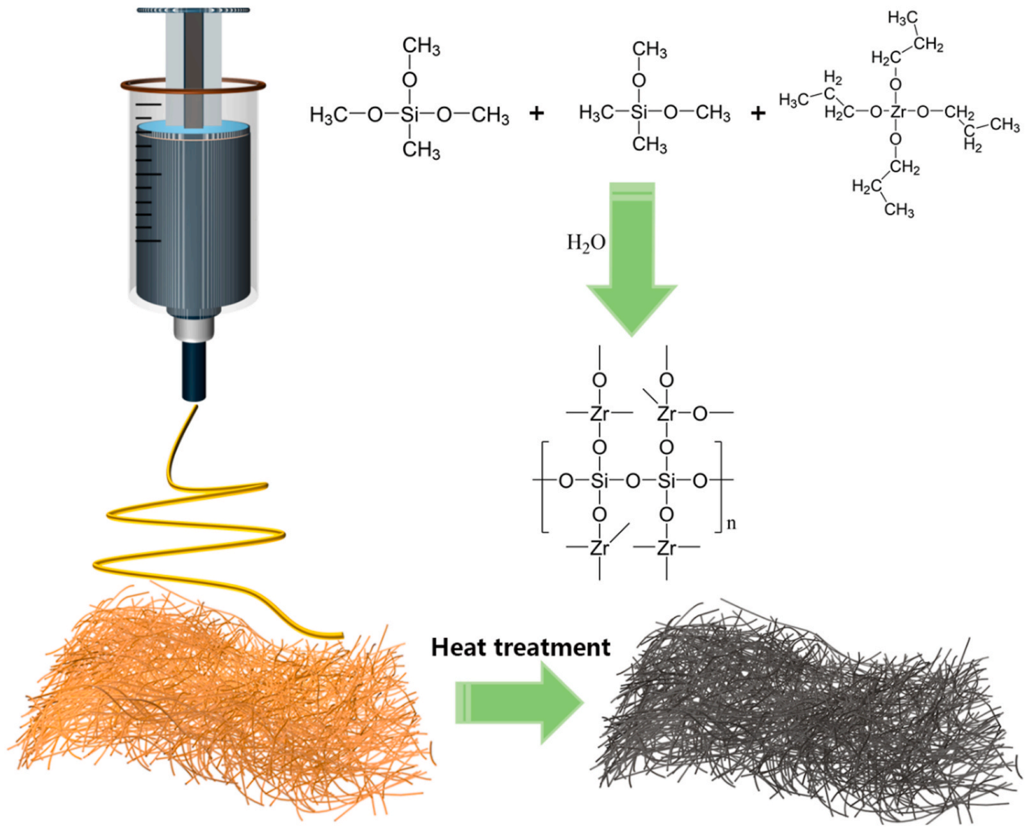

Fig. 1. Schematic illustration of the fabrication process of amorphous SiZrOC nanofibers.

strips. These studies highlight that compositional regulation can effective balance impedance matching and dielectric loss, offering a promising path toward high-performance absorbers.

Besides composition tailoring, structural engineering plays a vital role in modulating EM responses [30–32]. Hierarchical structures such as porous frameworks, hollow spheres, core-shell architectures, and nanofibers have been shown to improve microwave attenuation by promoting multiple scattering and interfacial polarization [33–37]. Among these, electrospinning provides a versatile and scalable technique to fabricate continuous one-dimensional nanofibers with high aspect rations, tunable pore porosity, and large surface-to-volume ratios [38–40]. These features enhance microwave transmission pathway and impedance matching, make electrospun nanofibers promising candidates for lightweight, high-performance MAMs. For example, Jiang et al. [41] fabricated a lightweight SiBCN/SiC nanowires aerogel, achieving an RL_{min} of −50.1 dB and a bandwidth of 6.4 GHz at just 2 mm thickness, attributed to improved impedance matching and multi-reflection effects. In a subsequent study, Jiang et al. [42] fabricated SiBCN/SiBCN nanofiber ceramic aerogel using electrospinning, reaching an RL_{min} of −54.24 dB and a bandwidth of 7.48 GHz, further demonstrating the potential of electrospun structures. In our previous work [43] reported SiOC nanospheres fabricated by the PDCs route with excellent microwave absorption properties (RL_{min} of −56.82 dB at 2.93 mm thickness). However, limited effects have been made to further enhance the absorption of SiOC ceramics by combining the introduction regulation (secondary phase introduction) with structure design (nanofibers).

In this work, one-dimensional SiZrOC nanofibers were fabricated by electrospinning of silazirconoxane precursors and followed by controlled pyrolysis under argon. The electrospun nanofibers proved continuous transmission pathway, while the in-situ formation of ZrO_2 nanoparticles within the amorphous matrix introduces interfacial polarization and multiple scattering centers. As a results, the SiZrOC nanofibers exhibit excellent EM wave absorbing performance, highlighting potential for applications in advanced communications systems and extreme environments.

2. Experimental methods

2.1. Materials

Methyltrimethoxysilane ($CH_3Si(CH_3O)_3$), dimethyldimethoxysilane ($(CH_3)_2Si(CH_3O)_2$), polyvinyl pyrrolidone (PVP), N,N-dimethylformamide (DMF), and zirconium acetylacetonate ($Zr(acac)_4$) were purchased from Shanghai McLean Biochemical Technology Co., Ltd. All chemicals were used as received without further purification.

2.2. Preparation of amorphous SiZrOC nanofibers

Amorphous SiZrOC nanofibers were fabricated via electrospinning followed by pyrolysis at different temperatures. The precursor solution was obtained by homogeneously mixing methyltrimethoxysilane, dimethyldimethoxysilane, $Zr(acac)_4$, PVP, DMF, H_2O at a weight ratio of 5:1:3:1:1 under magnetic stirring at room temperature for 8 h. The resulting solution was electrospun using a spinning apparatus (E02, Foshan Weimai Technology Co., Ltd., China) onto an oil-coated paper collector (Fig. 1). The electrospinning parameters were set as follows: tip-to-collector distance, 15 cm; applied voltage, 20 kV; and feeding rate, 0.3 ml/h. The ambient temperature and relative humidity were maintained at ~28 °C and ~60 %, respectively. After electrospinning, the obtained white precursor nanofibers were cured in air at room temperature for 24 h. Subsequently, they were pyrolyzed at different temperatures (800–1000 °C) in argon atmosphere for 1 h at a heating rate of 2 °C/min, yielding amorphous SiZrOC nanofibers.

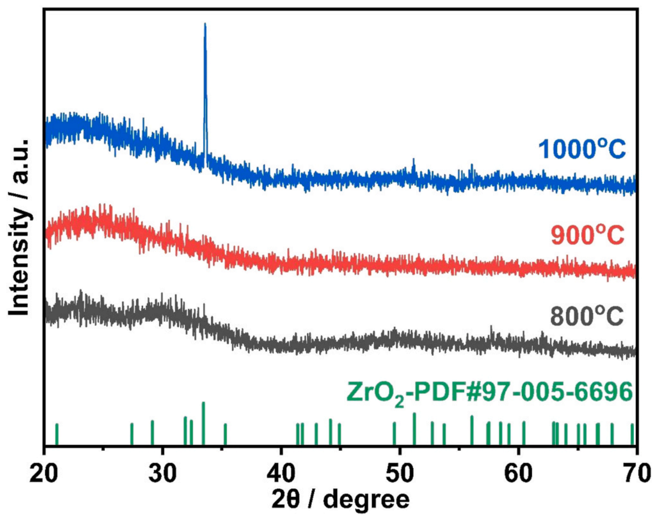

Fig. 2. XRD patterns of SiZrOC nanofibers pyrolyzed at different temperatures.

2.3. Characterization

The phase composition was analyzed by X-ray diffraction (XRD, Ultima IV, Rigaku, Japan) using Cu Kα radiation over 2θ range of 5–90°. The morphologies and microstructures were examined by scanning electron microscopy (SEM, JSM-IT800, JEOL, Japan) and transmission electron microscopy (TEM, CM300, Philips, Netherlands), both equipped with energy-dispersive X-ray spectroscopy (EDS). Elemental compositions and chemical states were determined by X-ray photoelectron spectroscopy (XPS, K-Alpha, Thermo Scientific, USA).

The EM properties were measured using the transmission/reflection (T/R) coaxial line method on a vector network analyzer (N5230A, Agilent, USA) in the frequency range of 2–18 GHz. For measurement, SiZrOC nanofibers were uniformly mixed with paraffin (30–50 wt% powder content) and pressed into cylindrical rings with an inner diameter of 3.04 mm, outer diameter of 7.00 mm, and thickness of 2.0–4.0 mm.

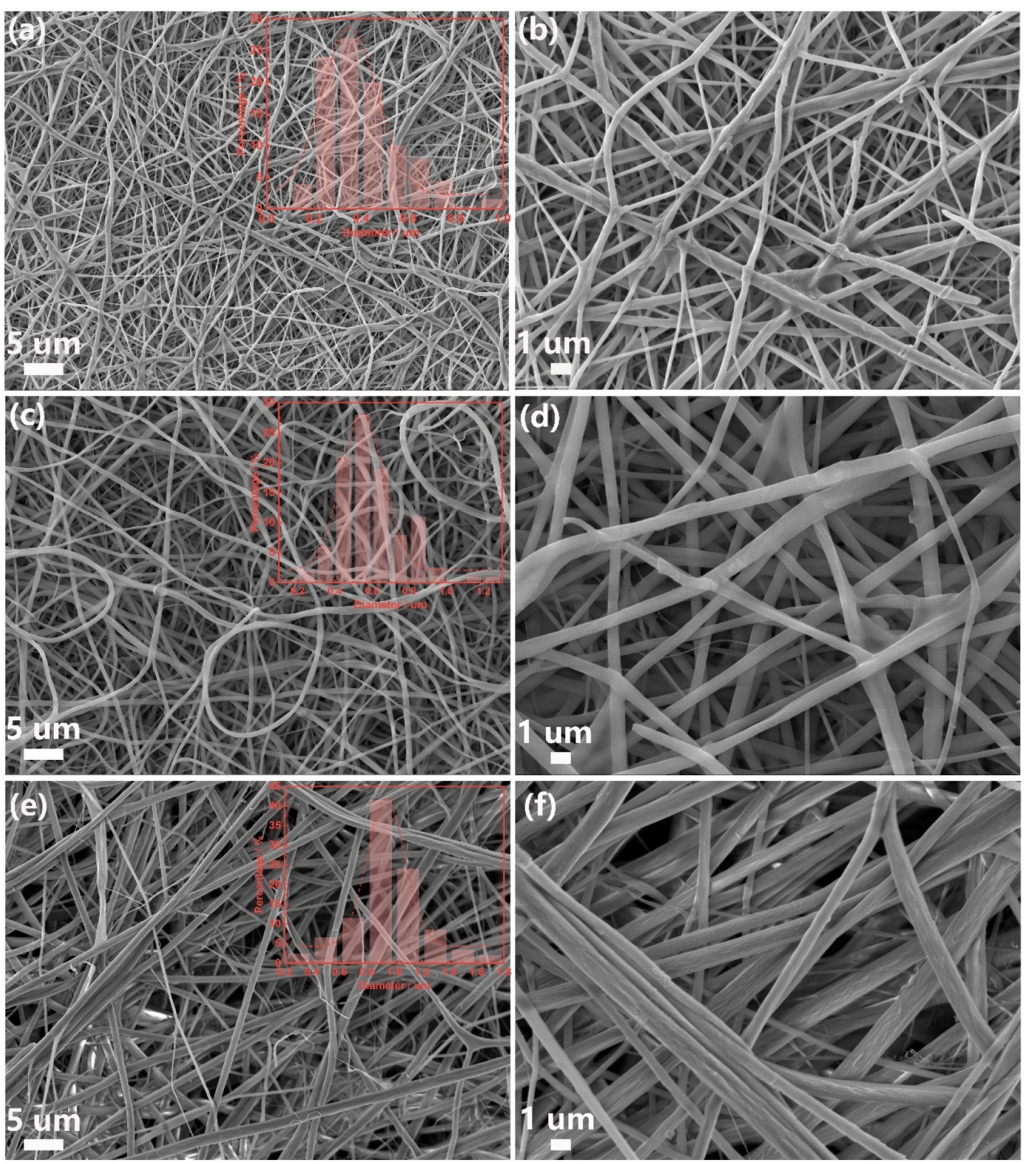

Fig. 3. SEM images of amorphous SiZrOC nanofibers pyrolyzed at different temperatures (a, b) 800 °C; (c, d) 900 °C; (e, f) 1000 °C.

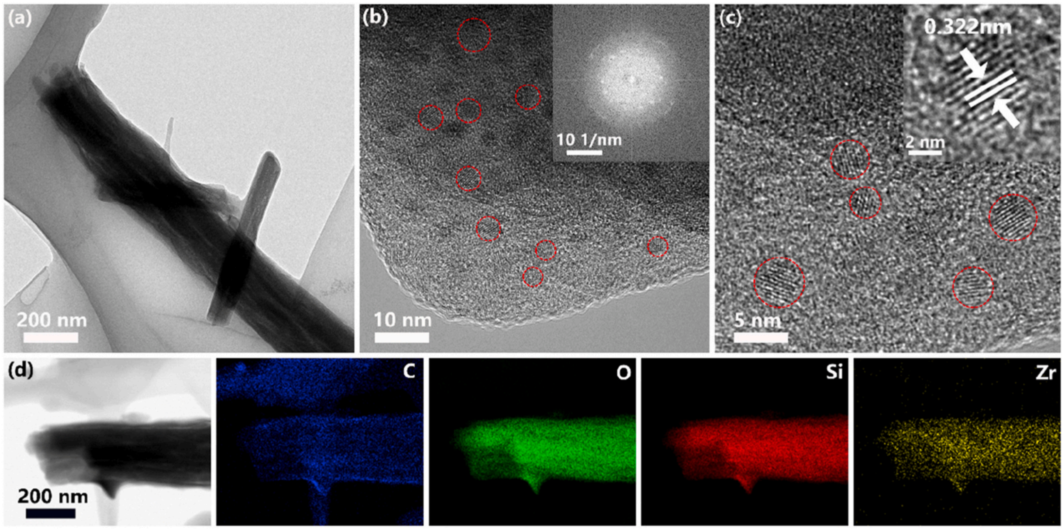

Fig. 4. Microstructure of the amorphous SiZrOC nanofibers heat-treated at 1000 °C: (a) TEM image; (b, c) HRTEM images and SAED pattern: nanocrystals within the amorphous matrix; (d) elemental mappings of Si, Zr, O, and C.

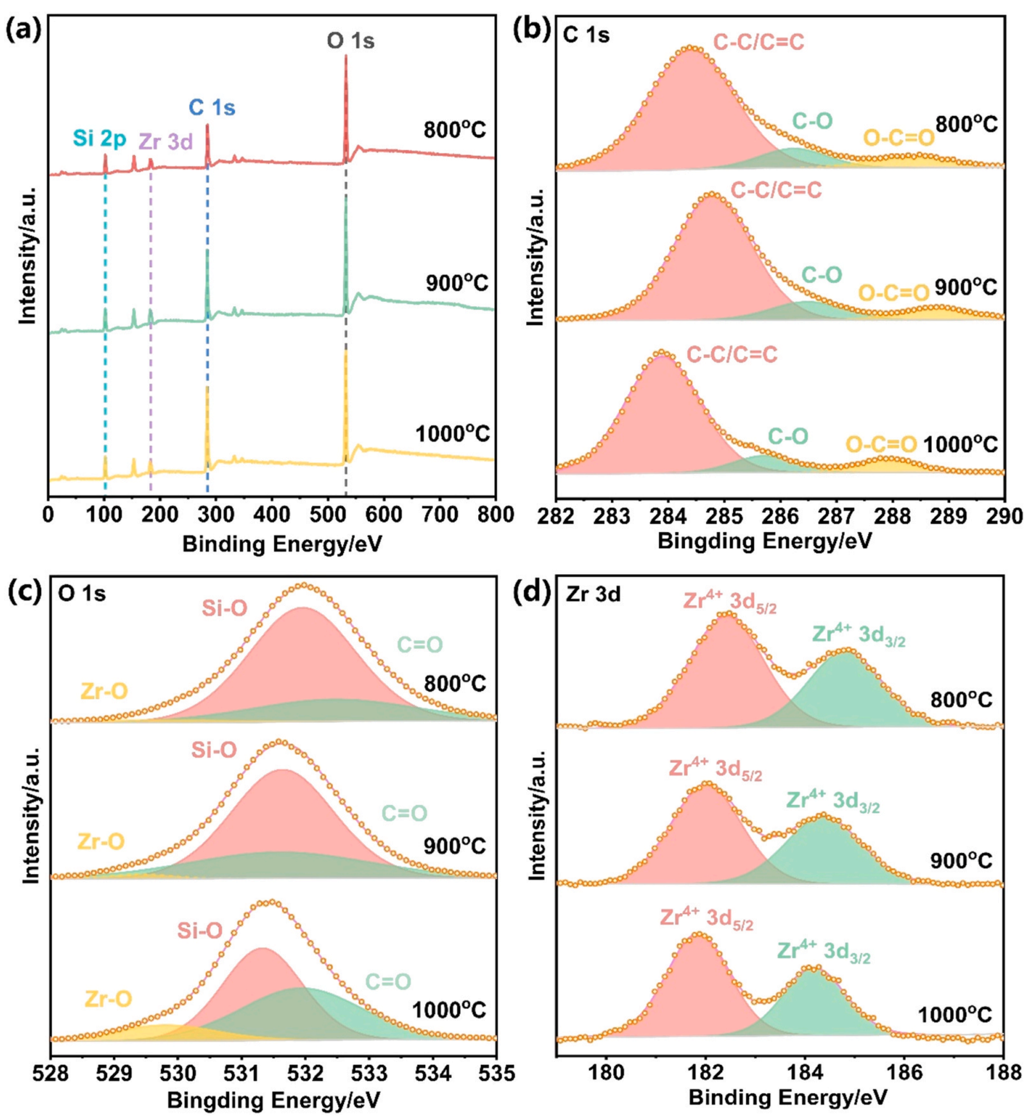

Fig. 5. XPS spectra of SiZrOC nanofibers: (a) wide-scan XPS spectra; high-resolution spectra of (b) C1s, (c) O1s, and (d) Zr 3d.

3. Results and discussion

3.1. Phase and morphology of the amorphous SiZrOC nanofibers

Fig. 2 shows the XRD patterns of SiZrOC nanofibers pyrolyzed at different temperatures. At lower pyrolysis temperatures, the broad diffraction hump suggests a predominantly amorphous structure, typical of PDCs at lower pyrolysis temperatures. With increasing pyrolysis temperature, distinct diffraction peaks gradually emerge, corresponding to the crystallization of phase, ZrO_2. When the temperature reaches 1000 °C, a sharp peak appears at ~34°, accompanied by additional weaker peaks at around 28° and 50°, which can be indexed to the ZrO_2 phase. These results, consistent with previous reports [39], suggest that phase separation occurs during pyrolysis, leading to the gradual formation of ZrO_2 within the amorphous structure.

Fig. 3 illustrates representative SEM images of SiZrOC nanofibers pyrolyzed at different temperatures, highlighting their temperature-dependent morphological evolution. At 800 °C the fibers show an averages diameter of approximately 360 nm and retain a uniform amorphous morphology characterized by smooth surface and a well-interconnected network. When the pyrolysis temperature is increased to 900 °C, the average fiber diameter expands to about 550 nm. At this stage, partial fiber fusion and a noticeable increase in surface roughness emerge, which can be attributed to thermally activated structural rearrangement and the onset of localized sintering. With further temperature elevation (e.g., ≥1000 °C), the fiber diameter increases to approximately 940 nm. This coarsening is accompanied by more pronounced surface roughening, consistent with crystallization and phase-separation phenomena commonly observed in polymer-derived ceramic systems. These progressive structural transformations strongly influence the microstructural integrity and, consequently, the functional performance of the nanofibers.

3.2. Microstructure and chemical characteristic of the amorphous SiZrOC nanofibers

TEM analysis was carried out to further confirm the structural characteristics of the amorphous SiZrOC nanofibers. As shown in Fig. 4a, the fibers exhibit diameters of approximately 200 nm, with some thinner fibers measuring around 100 nm, in good agreement with the SEM observations. High-resolution TEM (HRTEM) reveals numerous nanocrystalline domains (Fig. 4b and c) embedded within the amorphous matrix, forming abundant heterogeneous interfaces. The SAED pattern (Fig. 4b) exhibits a dominate amorphous ring due to the bulk amorphous SiOC matrix, but also contains weak and diffuse diffraction spots originating from the nanometer-scale ZrO_2 crystallites. Such coexistence-an amorphous halo with superimposed faint nanocrystal reflections-is characteristic of PDC undergoing early-stage phase separation and nanocrystallization. The lattice fringes observed in Fig. 4c show a measured interplanar spacing of ~0.322 nm, which is consistent with the (020) planes spacing of orthorhombic ZrO_2. Elemental mappings show the uniform distribution of C, O, Si, and Zr, validating the successful fabrication of SiZrOC nanofibers. This hybrid nanostructure, comprising nanocrystals embedded in an amorphous matrix, is advantageous for EM wave attenuation. On one hand, the amorphous matrix contains a high density of defects of structural defects; on the other hand, the abundant heterogeneous interfaces between the nanocrystals and the matrix promote interface polarization. These synergistic factors effectively promote dielectric loss.

The chemical bonding states of the SiZrOC nanofibers were investigated by XPS (Fig. 5). The wide-scan XPS spectra (Fig. 5a) confirms the presence of Si, Zr, O, and C elements in fibers pyrolyzed at 800 °C, 900 °C, and 1000 °C. High-resolution spectra of C1s, O1s, and Zr3d provide further insights into the evolution of chemical states with pyrolysis temperature. The C1s spectrum (Fig. 5b) displays a major peak at ~284.6 eV, attributed to C-C/C=C bonds (graphitic carbon). Additional

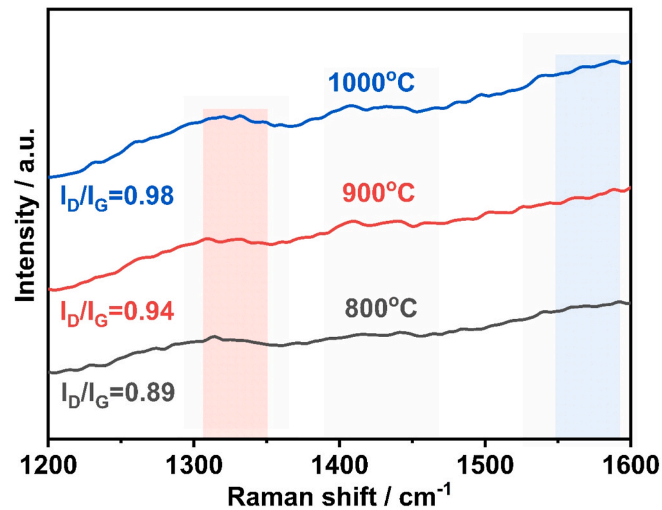

Fig. 6. Raman spectra of SiZrOC nanofibers pyrolyzed at different temperatures.

peaks at around 286.2 eV and 288.5 eV correspond to C-O and O-C=O species, respectively. With increasing pyrolysis temperature, the intensity of oxygenated carbon peaks decreases, indicating the decomposition of organic residues and purification of the carbon phase. The O1s spectra (Fig. 5c) shows two peaks at ~531.3 eV, associated to Si-O and Zr-O bonds, respectively, along with a weaker peak at around 532.4 eV, corresponding to C=O groups. Increasing pyrolysis temperature leads to a stronger inorganic oxygen signal and weaker high-binding-energy components, suggesting the growth of oxide networks. The Zr3d spectra (Fig. 5d) exhibit two peaks at ~182.3 eV and ~184.7 eV, corresponding to Zr $3d_{5/2}$ and Zr $3d_{3/2}$ of Zr^{4+} in ZrO_2. The gradual enhancement and slight shift of these peaks toward higher binding energies with increasing temperature reflect progressive oxidation and structural ordering of the Zr-based network. Collectively, these results indicate that increasing the pyrolysis temperature drives the transformation from an organic-rich precursor to a hybrid nanostructure composite of Zr-O inorganic frameworks and partially graphitized carbon. This structural evolution provides synergistic regulation of conductive, dielectric, and interfacial components, resulting in optimized microwave absorption. Raman spectra further confirm the evolution of the carbon phase. As can be seen from Fig. 6, when the temperature increases from 800 °C to 1000 °C, two prominent peaks gradually appear approximately 1330 cm^{-1} and 1570 cm^{-1} for the SiZrOC fibers, corresponding to the disordered (D) and graphitization (G) peaks, respectively. The intensities of both D and G peaks increase with pyrolysis temperature, which signifies the gradual transformation of amorphous carbon into ordered graphitic domains. The I_D/I_G values for 800 °C, 900 °C, and 1000 °C are 0.89, 0.94, and 0.98, respectively. The gradual increase in I_D/I_G ratio with temperature indicates the presence of more defect structures and increased disorder with emerging graphitic ordering in the SiZrOC fiber matrix pyrolyzed at 1000 °C. Such defect-rich carbon structures are known to act as effective polarization centers, facilitating dipole polarization, hopping conduction, and defect-induced relaxation process.

3.3. EM wave absorbing performance of the SiZrOC nanofibers

The complex permittivity ($\varepsilon = \varepsilon' - j\varepsilon''$) of SiZrOC nanofibers was measured to evaluate their EM response over the 2–18 GHz frequency range. Fig. 7 presents the real (ε') and imaginary (ε'') parts of permittivity for nanofibers pyrolyzed at 800 °C (a and b), 900 °C (c and d), and 1000 °C (e and f) with varying fiber contents (30 wt%, 40 wt%, and 50 wt% in paraffin). At 800 °C, the nanofibers show relatively high ε' and ε'' values, particularly for the 50 wt% samples, with $\varepsilon' > 6$ and $\varepsilon'' > 2.0$

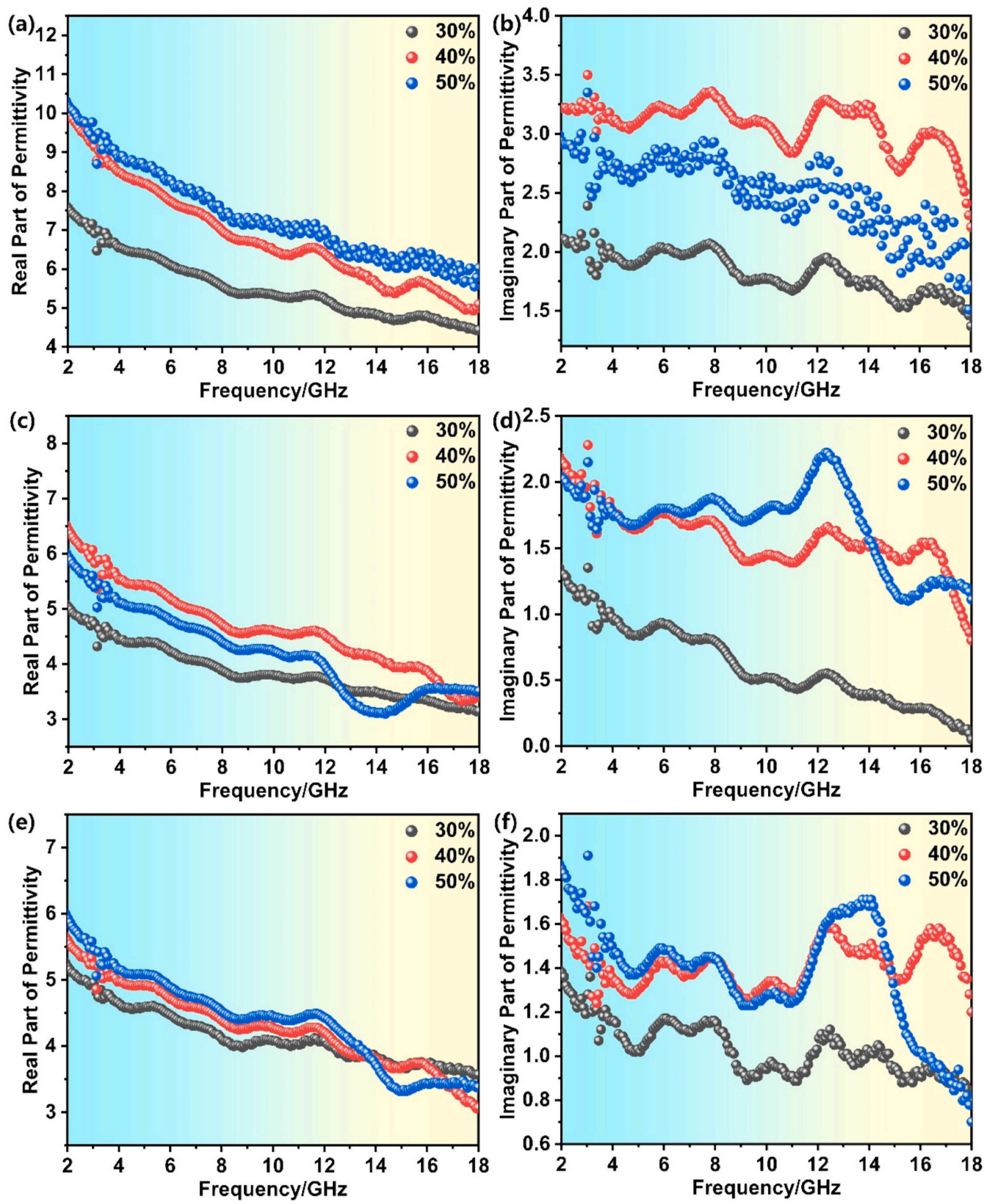

Fig. 7. EM properties of SiZrOC nanofiber at different pyrolysis temperature and fiber contents: (a, b) 800 °C, (c, d) 900 °C, and (e, f) 1000 °C; (a, c, e) real part of permittivity (ε'), (b, d, f) imaginary part of permittivity (ε'').

across much of the frequency range, indicating strong polarization capability and stronger dielectric loss. Both ε' and ε'' gradually decrease with increasing frequency, attributed to weakened dipole orientation polarization. For nanofibers pyrolyzed at 900 °C, ε' and ε'' decrease slightly compared to the 800 °C sample, indicating that partial crystallization or phase transformation begins to reduce the complex permittivity. Notably, the ε'' values of 50 wt% samples remain above 1.5 across most the Ku-band, suggesting sustained dielectric loss. At 1000 °C, both ε' and ε'' reach their lowest values among all samples, with ε' dropping below 5 for the 30 wt% fibers and ε'' generally below 1.5. This decrease is attributed to extensive formation of ZrO_2 phases, which possess relatively low dielectric constants. At a fixed pyrolysis temperature, increasing fiber content from 30 to 50 wt% enhances both ε' and ε'' values, likely due to the formation of more continuous conductive or semi-conductive networks within the paraffin matrix.

Fig. 8 shows the dielectric loss angular tangent (tan $\delta = \varepsilon''/\varepsilon'$), reflecting the material's ability to dissipate EM energy. The 50 wt% sample pyrolyzed at 1000 °C exhibits significantly higher tan δ compared to other lower-temperature samples (Fig. 8e), resulting from amorphous phases and defects structures that promote interfacial and dipolar relaxation. For fiber pyrolyzed at 800 °C and 900 °C, tan δ fluctuates between 0.2 and 0.6 (Fig. 8a and c), indicating enhanced conductivity due to higher degree of graphitization. At a fixed temperature, increasing fiber content generally increase tan δ, although an exception is observed for the 40 wt% sample at 800 °C. Given that the material is essentially non-magnetic, dielectric loss dominates the absorption mechanism. The dielectric response can be described by Debye's theory, where ε' and ε'' are descripted as follows [44–46]:

$$\varepsilon' = \varepsilon_\infty + \frac{\varepsilon_s - \varepsilon_\infty}{1 + (2\pi f \tau)^2} \tag{1}$$

$$\varepsilon'' = \frac{2\pi f \tau(\varepsilon_s - \varepsilon_\infty)}{1 + (2\pi f \tau)^2} + \frac{\sigma}{2\pi f \varepsilon_0} \tag{2}$$

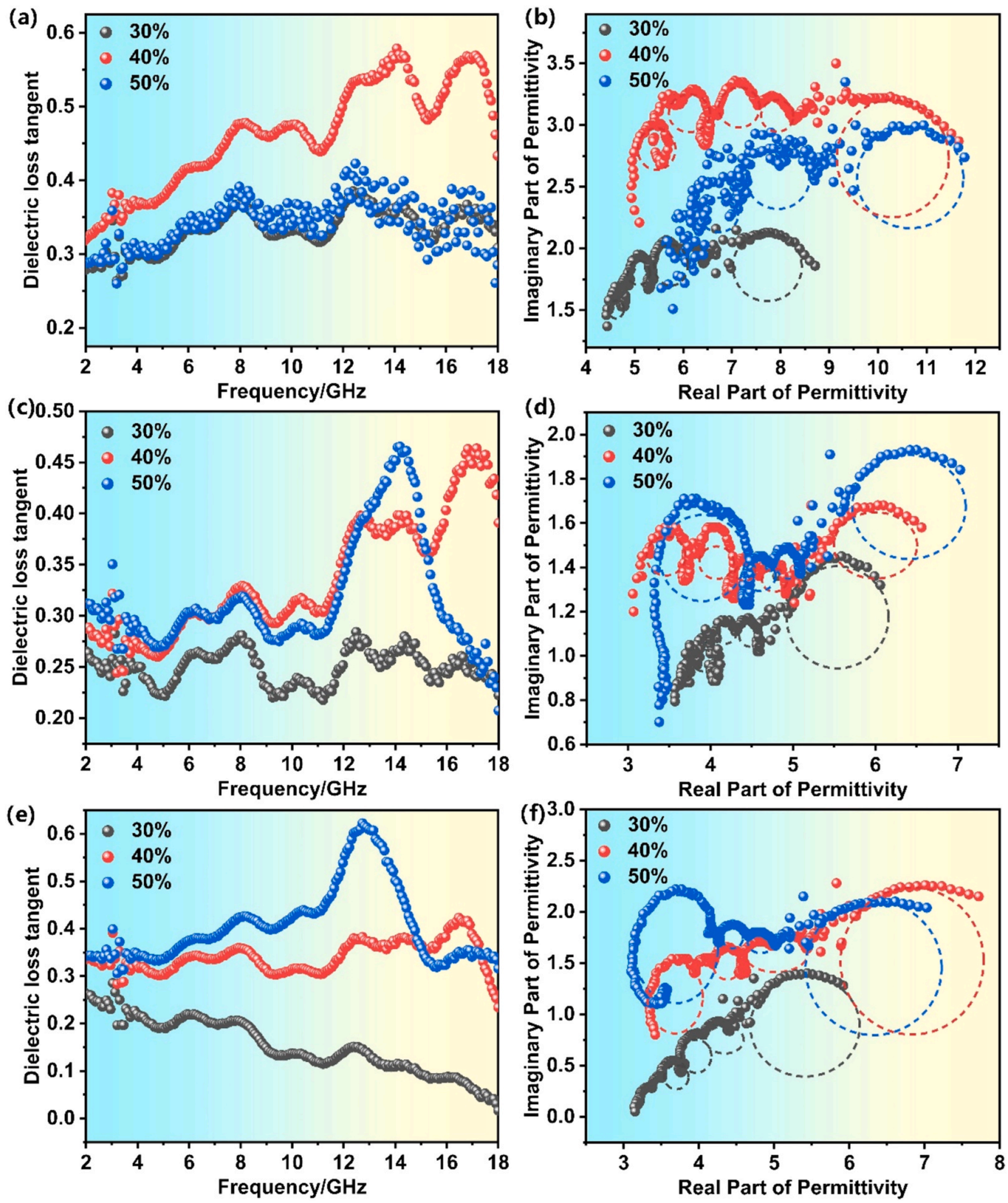

Fig. 8. Dielectric loss tangents and Cole-Cole plots of SiZrOC amorphous nanofibers pyrolyzed at different temperatures: (a, c, e) dielectric loss tangent (tan δ), (b, d, f) Cole-Cole semicircles; (a, b) 800 °C, (c, d) 900 °C, (e, f) 1000 °C.

where ε_∞ and ε_s are the high-frequency and static permittivity, f is the frequency; τ is the relaxation time, and σ is the conductivity of the sample. In Equation (2), the first term represents the polarization relaxation, while the second term accounts for the conduction. The relationship between of ε' and ε'' can be further described as [47–49]:

$$\left(\varepsilon' - \frac{\varepsilon_s + \varepsilon_\infty}{2}\right)^2 + (\varepsilon'')^2 = \left(\frac{\varepsilon_s - \varepsilon_\infty}{2}\right)^2 \tag{3}$$

Equation (3) produces semicircular ε'-ε'' curves, known as Cole-Cole plots, representing Debye relaxation process. Fig. 8b, d, and f show the Cole-Cole plots of SiZrOC nanofibers. Multiple semicircles are observed for all the samples, indicating the coexistence of multiple polarizations mechanisms contributing to relaxation loss. The tails at low frequencies confirm the presence of conduction loss. Higher pyrolysis temperatures enhance both conductive loss and polarization loss, attributed to increasing graphitization (Ohmic loss) and interfacial polarization at defect sites and ZrO_2 interface. Therefore, the dielectric and conduction loss dominate the EM wave absorption of the amorphous SiZrOC nanofibers.

According to transmission line theory, the EM wave absorption of SiZrOC nanofibers can be evaluated by reflection loss (*RL*), which is calculated from the EM parameters as follows [50,51]:

$$RL(dB) = 20 \log\left|\frac{Z_{in} - Z_0}{Z_{in} + Z_0}\right| \tag{4}$$

$$Z_{in} = Z_0\sqrt{\frac{\mu_r}{\varepsilon_r}}tanh\left[j\left(\frac{2\pi fd}{c}\right)\sqrt{\mu_r\varepsilon_r}\right] \tag{5}$$

where Z_{in} and Z_0 are the input impedance of the material and the free space impedance, respectively; d is the absorber thickness and c is the speed of light in vacuum. Usually, $RL < 10$ dB suggest that over 90 % of the incident EM wave is absorbed, and the corresponding frequency range is defined as the effective absorption bandwidth (EAB). Fig. 9 displays the RL of SiZrOC nanofibers pyrolyzed at different temperatures

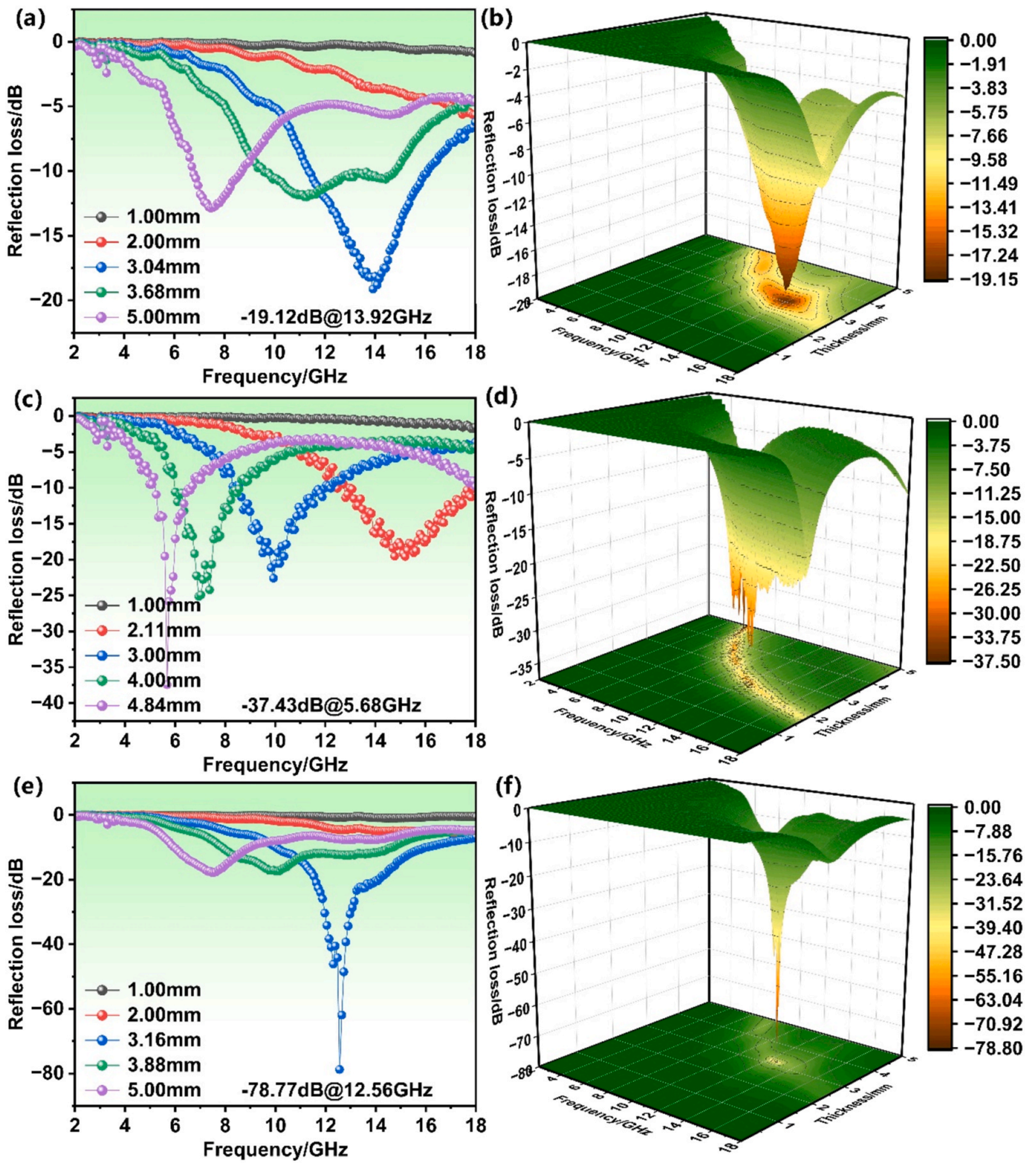

Fig. 9. The two-dimensional and three-dimensional reflection loss (*RL*) of SiZrOC nanofibers (50 wt% content) as a function of frequency and thickness: (a, b) pyrolysis at 800 °C; (c, d) pyrolysis at 900 °C; (e, f) pyrolysis at 1000 °C.

with 50 wt% fiber content. The others show in Figs. S1–S2 (Supporting information). As depicted in Fig. 9e and f, the sample pyrolyzed at 1000 °C exhibits an impressive RL_{min} of −78.77 dB at 12.56 GHz with a thickness of 3.16 mm and an EAB_{max} was 6.8 GHz at a thickness of 3.88 mm, revealing its optimal EM wave absorption performance. This excellent performance is attributed to numerous interfaces between the amorphous matrix and ZrO_2 nanocrystals, which enhance interfacial polarization and dielectric loss. By comparison, the samples pyrolyzed at 800 °C and 900 °C show RL_{min} value of −19.12 dB at 13.92 GHz and −37.43 dB at 5.68 GHz (Fig. 9a–d), respectively, which are higher due to fewer interfaces and reduced interfacial polarization. Notably, the 900 °C sample with 40 wt% fiber (Fig. S2, supporting information) content also exhibits excellent absorption, with RL_{min} value of −60.23 dB at 11.54 GHz with the thickness of 2.62 mm, and an EAB_{max} value of 5.2 GHz at the thickness of 6.89 mm (Table S1, supporting information). These results confirm that EM absorption strongly dependent on the pyrolysis temperatures, as higher temperature promote the formation of more nanocrystals, increasing interfacial polarization, relaxation polarization loss, and multiple scattering.

Dielectric properties are the primary factor in EM absorption, but impedance matching ($Z = Z_{in}/Z_0$) and attenuation constant (α) are also crucial. Impedance matching determines whether EM waves can effectively enter the material rather than being reflected at the surface. Ideal impedance matching occurs when $Z \approx 1$, meaning the material impedance equals free space impedance. The attenuation coefficient α determines the ability of the material to dissipate EM waves that have entered, and is calculated as [52–54]:

$$\alpha = \frac{2\pi f}{c} \times \sqrt{(\mu''\varepsilon'' - \mu'\varepsilon') + \sqrt{(\mu''\varepsilon'' - \mu'\varepsilon')^2 + (\mu''\varepsilon'' + \mu'\varepsilon')^2}} \quad (6)$$

Fig. 10a–c displays the impedance matching maps for 50 wt% SiZrOC nanofibers at different pyrolysis temperatures. In the 4–10 GHz range, a thickness of 3–5 mm allows the Z to approach 1, while in the 10–18 GHz range, a thickness of 1–3 mm provides ideal matching. The 1000 °C sample achieves RL_{min} = −78.77 dB at 12.56 GHz with a thickness of 3.16 mm, coinciding with the best impedance matching. Fig. 10d shows the attenuation constant α, which increase with frequency for all samples, indicating strong EM wave attenuation. The 1000 °C sample reaches a maximum αof 176.55 around 12 GHz, demonstrating that the combination of excellent impedance matching and high α enables

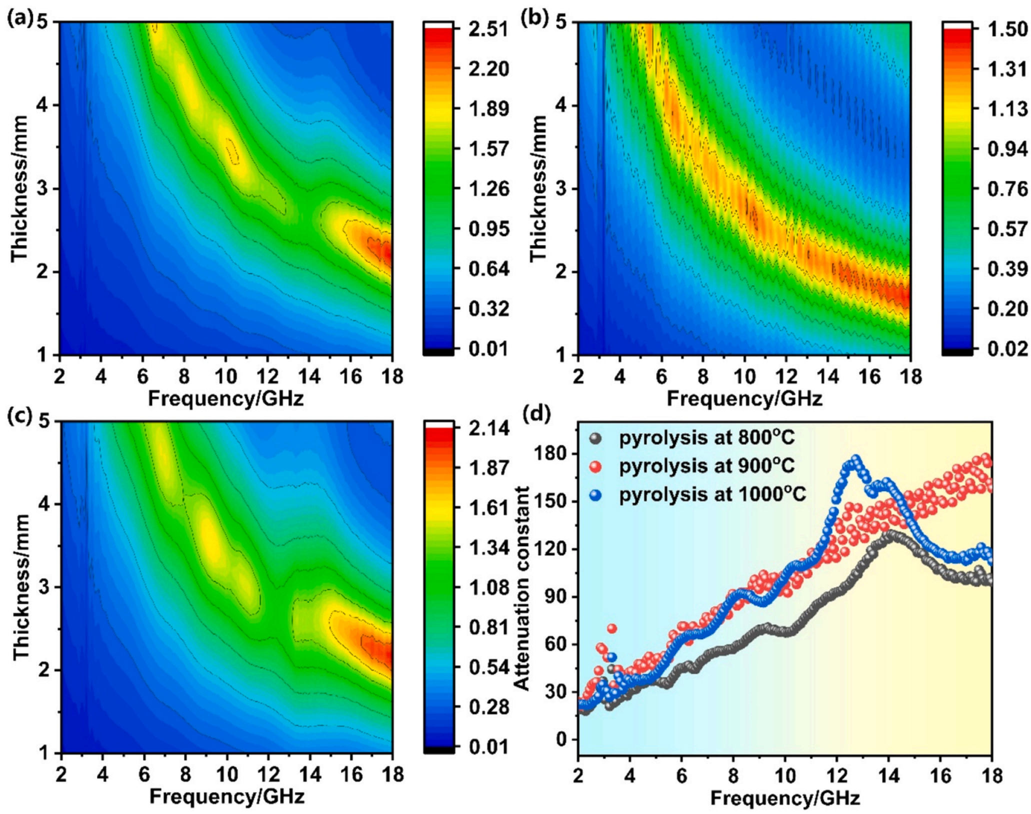

Fig. 10. Impedance matching maps and attenuation constant (α) of SiZrOC nanofibers (50 wt% content) pyrolyzed at different temperatures: (a–c) impedance matching maps for 800 °C, 900 °C, and 1000 °C, respectively; (d) attenuation constant (α).

superior absorption performance.

3.4. Microwave absorption mechanism of the SiZrOC nanofibers

The dielectric loss of different samples was fitted using the least squares method, as shown in Fig. 11a and b. The conduction loss is inversely proportional to frequency and directly proportional to conductivity (Fig. 11a). Meanwhile, higher pyrolysis temperatures enhance the conduction loss of SiZrOC nanofibers. From Fig. 11b, it can be observed that the polarization loss of SiZrOC nanofibers decreases with increasing pyrolysis temperature. This is because as the temperature rises, the microstructure coarsens, oxygen-containing functional groups gradually decompose, and ZrO_2 nanocrystals precipitate. These changes reduce the number of effective dipoles and interfacial polarization centers. At the same time, the increasing graphitization degree of the free carbon phase enhances electrical conduction, causing part of the dielectric loss mechanism to shift from polarization loss to conductive loss. The results indicate that conduction loss and polarization loss dominate the EM wave attenuation of the 1000 °C pyrolyzed sample, primarily due to its unique structure. The main mechanisms of EM wave absorption in SiZrOC nanofibers (50 wt%) pyrolyzed at 1000 °C can be understood from both microscopic and macroscopic perspectives, as shown in Fig. 11c. At the microscopic level, defects such as vacancy, dangling bonds, and the residual active functional sites in the SiZrOC nanofibers act as polarization centers under the alternating electromagnetic field, resulting in dipole polarization loss and enhanced electromagnetic filed, resulting in dipolar polarization loss and enhanced absorption. Additionally, in the SiZrOC nanofibers (50 wt%) pyrolyzed at 1000 °C, the amorphous matrix and numerous in-situ formed ZrO_2 nanoparticles generated abundant heterogeneous interfaces. These interfaces facilitate charge accumulation, leading to strong space charge polarization and increased interfacial loss. Higher pyrolysis temperatures promote the formation of free carbon, enhancing the electrical conductivity of the material. The increased conductivity facilitates carrier migration and hopping, inducing field-driven microcurrents and further enhancing conduction loss. From a macroscopic point of view, the highly porous structures of the SiZrOC nanofibers (50 wt%) pyrolyzed at 1000 °C provides multiple transmission pathways for incident EM wave, improves impedance matching, and promotes multiple reflections. Therefore, under the synergistic effects of interfacial polarization, dipole polarization, conduction loss, and multiple reflections, the SiZrOC nanofibers (50 wt%) pyrolyzed at 1000 °C exhibit outstanding microwave-absorbing properties.

The microwave absorption performance of SiZrOC nanofibers compared with other representative materials prepared via electrospinning are summarized in Table 1. The results clearly demonstrate that SiZrOC nanofibers exhibit a significantly lower RL_{min} value and a relatively broader EAB_{max}, outperforming most reported electrospun counterparts. In summary, the SiZrOC nanofiber materials display excellent microwave absorption performance, highlighting their potential in the field of microwave absorption.

4. Conclusion

In this study, SiZrOC nanofibers were successfully fabricated via electrospinning followed by controlled pyrolysis. By tuning the pyrolysis temperature, both the microstructure and EM properties of the samples can be effectively optimized, leading to improved impedance matching and enhanced attenuation capability. The sample pyrolysis at 1000 °C, consisting primarily of ZrO_2 nanoparticles embedded in an amorphous matrix, exhibits exceptional EM wave absorption properties due to the synergistic effects of its unique microstructure, enhanced conductive loss and polarization loss, and the macroscopic porous architecture. Notably, this sample achieves a minimum reflection (*RL*) value of −78.77 dB at 12.56 GHz with a thickness of 3.16 mm and a maximum effective absorption bandwidth was 6.8 GHz at a thickness of 3.88 mm.

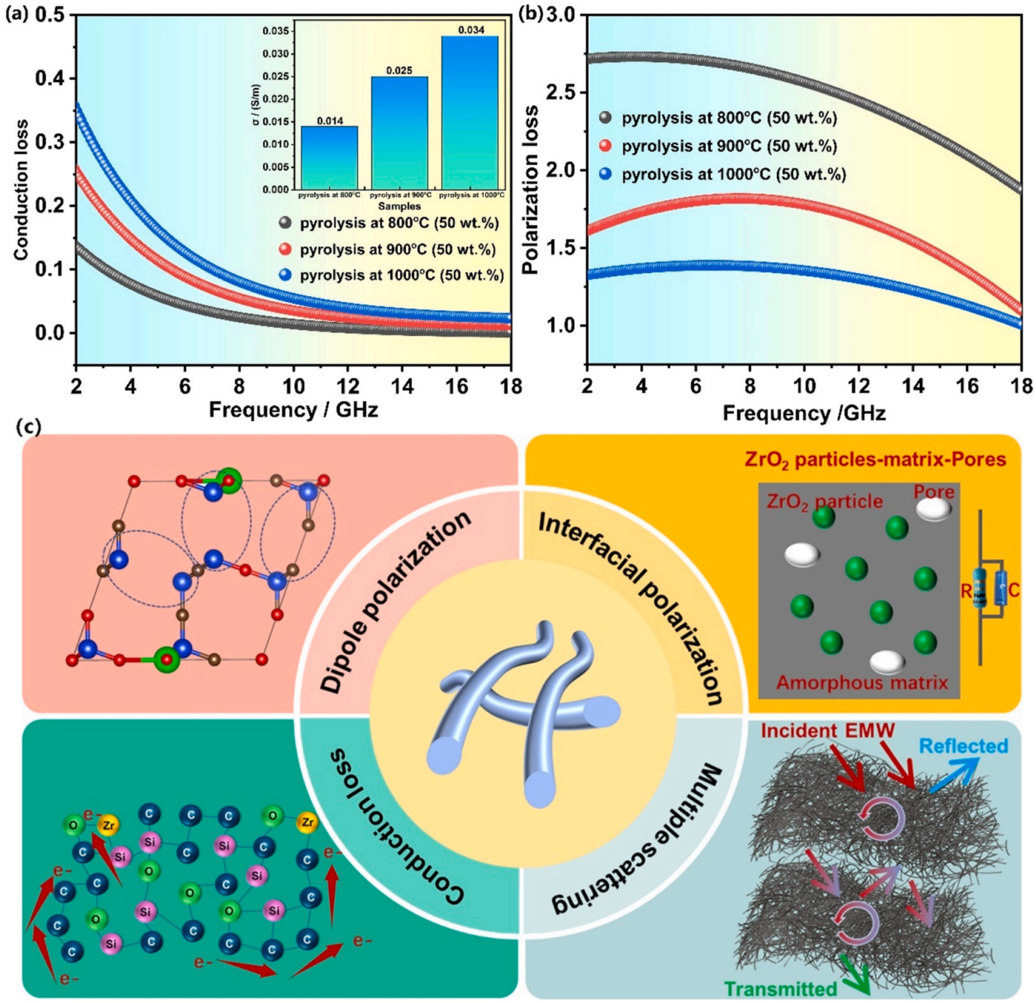

Fig. 11. Schematic illustrations of the EM wave absorption mechanisms of SiZrOC nanofibers (50 wt%) pyrolyzed at 1000 °C: (a) conduction loss; (b) polarization loss, and (c) overall absorption mechanism.

Table 1
Comparison of microwave-absorbing performance among SiZrOC nanofibers and other representative materials prepared via electrospinning.

Material	Thickness/mm	RL_{min}/dB	EAB_{max}/GHz	Ref.
MFO/CGFS-3	4.8	−57.21	3.58	[2]
$Fe_{0.64}Ni_{0.36}$/MXene/CNFs	2.7	−54.1	7.76	[4]
Fe_3O_4@NRC	1.5	−37.4	4.16	[5]
Fe/Co@C-CNFs	1.08	−18.66	4.2	[7]
SiC/C	3.0	−36	4.1	[9]
HCSS/Zr/SiC	3.5	−48.6	3.2	[11]
ZrO_2/CF–rGO	3.14	−62.99	8.19	[30]
ZS-HNFA-x	2.4	−53.2	6.4	[33]
ZrO_2/ZrB_2/C	4.0	−54	3.1	[35]
ZrO_2/ZrC/ZrB_2	2.4	−21	8.64	[37]
SiCN	2.0	−31.1	4.59	[40]
CeO_2/N-C	2.5	−42.95	8.48	[55]
Ni PNFs	2.5	−40.4	4.6	[56]
SiZrOC	3.16	−78.77	6.8	This work

This work demonstrates, for the first time, the significant contribution of a crystal-amorphous hybrid structure to the dielectric loss in SiZrOC nanofibers and provides a valuable guideline for designing highly efficient electromagnetic wave absorbers.

CRediT authorship contribution statement

Saidi Wang: Writing – review & editing, Writing – original draft, Formal analysis, Data curation. **Haixuan Ou:** Funding acquisition, Formal analysis, Data curation. **Kaicheng Pei:** Funding acquisition, Formal analysis, Data curation. **Yangxuan Ou:** Funding acquisition, Formal analysis, Data curation. **Xiaoang Liu:** Funding acquisition, Formal analysis, Data curation. **Yimin Ouyang:** Funding acquisition, Formal analysis, Data curation. **Weiwei Xing:** Funding acquisition, Formal analysis, Data curation. **Bin Du:** Writing – review & editing, Writing – original draft, Resources, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

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